Lab2:
C++ programming with UNIX

EC327

“Introduction to Software Engineering”

Spring 2020

LAB1: SUMMARY

- Last's lab went over commands such as the following:

```
ls
cd
mv
man
:q
zip
```

- These commands and more are on the "Unix Cheat Sheet" that is uploaded to blackboard

AGENDA

- **The GNU C++ compiler (`g++`)**

- **UNIX**
  - **Environment Variables**
  - **Piping**
  - **New commands:**
    - `cat`, `less`, `tail`, `more`

- **User Input**

The GNU g++ Compiler

hello.cpp:

```
#include <iostream>

using namespace std;

int main()
{
    cout << "Hello World!" << endl;
    return 0;
}
```

The GNU g++ Compiler

Compile:

```
g++ hello.cpp
```

Execute:

```
./a.out
```

Environment Variables

```
echo $PATH
```

```
/opt/local/bin:/opt/local/sbin:/usr/bin:/bin
```

Directories of executables

The GNU g++ Compiler

Phases:

- **Preprocessing**
  - #include
  - #define

```
g++ -E hello.cpp

g++ -E hello.cpp | less

g++ -E hello.cpp | tail

g++ -E hello.cpp | head

g++ -E hello.cpp > hello.E
```

The GNU g++ Compiler

Phases:

- **Preprocessing**
  - #include
  - #define
- **Lexing and Parsing**
  - Syntax Checking
- **Compilation from C to Assembly to Machine Code**
- **Linking multiple Object Files**

```
g++ -S hello.cpp
g++ -c hello.cpp
cat hello.s
cat hello.o
```

Assembly Code

Object File

The GNU g++ Compiler

Phases:

- **Preprocessing**
  - #include
  - #define
- **Lexing and Parsing**
  - Syntax Checking
- **Compilation from C to Assembly to Machine Code**
- **Linking multiple Object Files**

```
g++ -E hello.cpp
g++ -S hello.cpp
g++ -c hello.cpp
```

Preprocessing
Assembly Code
Object File

```
g++ hello.cpp -o hello
./hello
```

Naming the Executable

User Input (1/2)

user_input.cpp

```
#include <iostream>
using namespace std;
int main(){
        // initializing number variable
        int number;
        // reading user input
        cin >> number;
        // output
        cout << ”Your input is ” << number << ”!” << endl;
        return 0;
}
```

User Input (2/2)

Compile:

```
g++ user_input.cpp -o lab2
```

Execute:

```
./lab2
```

Piping and Input/Output Redirection

words.txt:

```
Apple
Apricot
Avocado
Banana
Bilberry
Blackberry
Blackcurrant
Black sapote
Blueberry
Boysenberry
Cactus pear
Crab apple
Currant
Cherry
Chico fruit
Cloudberry
Coconut
Cranberry
```

Piping and Input/Output Redirection

- Piping
  - | : pipe character

```
cat words.txt | wc
ls | sort
```

- Output Redirection

```
echo ”Hello world!” > hello_world.txt
./hello_world > output.txt
```

Lab2: Takeaways

Re-compile when changing the source code

Compile all .cpp Files

Name the executable

```
g++ hello.cpp -o hello
```

Only clean code is good code!

```cpp
#include <iostream>
using namespace std;
int main()
{
cout<<"Bad Hello World!";
return 0;
}
```

BAD!

```cpp
#include <iostream>

using namespace std;

int main()
{
        cout << "Hello World!" << endl;

        return 0;
}
```

GOOD! ☺

SSH DEMO

QUESTIONS?